

December 2023

Duration: 3 Hours

Max. Marks: 100

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

20 Marks

- What is the primary function of a MOSFET in electronic circuits?**
 - To store data
 - To amplify signals
 - To generate heat
 - To provide mechanical support
- When a MOSFET is operating as a switch, in which region does it dissipate the least power?**
 - Cutoff region
 - Saturation region
 - Active region
 - Linear region
- In an N-channel enhancement-mode MOSFET, what happens when a positive voltage is applied to the gate terminal relative to the source terminal?**
 - The MOSFET turns off completely.
 - The channel becomes less conductive.
 - The source and drain terminals switch roles.
 - The channel becomes more conductive.
- In an inverting amplifier configuration using an op-amp, if the input voltage (V_{in}) is positive, what is the polarity of the output voltage (V_{out})?**
 - Positive
 - It depends on the gain.
 - Zero
 - Negative
- In a non-inverting op-amp configuration, if the input voltage (V_{in}) is positive, what is the phase relationship between V_{in} and the output voltage (V_{out})?**
 - V_{in} lags V_{out} by 90 degrees.
 - V_{in} leads V_{out} by 90 degrees
 - V_{in} and V_{out} are in phase.
 - V_{in} and V_{out} are 180 degrees out of phase.
- Slew rate is typically measured in which units?**
 - Hertz (Hz)
 - Volts per second (V/s)
 - Ohms (Ω)
 - Farads (F)
- A higher slew rate in an op-amp indicates:**
 - Slower response to input changes
 - Faster response to input changes
 - Greater power consumption
 - Lower voltage gain
- What is the primary purpose of a difference amplifier?**
 - To amplify the difference between two input signals
 - To amplify the sum of two input signals
 - To provide a single-ended output
 - To eliminate common-mode signals
- In a difference amplifier, the common-mode rejection ratio (CMRR) is a measure of:**
 - How well the amplifier amplifies the difference signal
 - How well the amplifier rejects common-mode signals
 - The total gain of the amplifier
 - The power supply voltage
- In a precision rectifier, what is the purpose of the diode(s) and operational amplifier?**
 - The diode(s) rectify the signal, and the op-amp provides gain.
 - The diode(s) provide gain, and the op-amp rectifies the signal.
 - Both the diode(s) and op-amp rectify the signal.
 - Neither the diode(s) nor op-amp are used in a precision rectifier.

11. In astable mode, a 555 timer generates a continuous square wave output. What are the components that determine the frequency of this square wave?
 - A) The inductor (L) and the resistor (R)
 - B) The voltage supply (Vcc) and the ground (GND)
 - C) The operational amplifier (op-amp) and the diode
 - D) The resistor (R) and the timing capacitor (C)
12. De Morgan's Second Theorem states that the complement of the sum of two variables is equal to:
 - A) The exclusive OR of their complements
 - B) The product of their complements
 - C) The sum of their complements
 - D) The inverse of their complements
13. Using De Morgan's Theorems, what is the complement of the expression $(A + B) \cdot C$?
 - A) $A' + B' + C'$
 - B) $A' \cdot B' \cdot C'$
 - C) $A' + B' \cdot C'$
 - D) $A' \cdot B' + C'$
14. How many half-adders are required to create a 4-bit binary adder circuit?
 - A) 1
 - B) 3
 - C) 2
 - D) 4
15. In a 4-to-1 multiplexer, how many select lines (control inputs) are required to select one of the four data inputs?
 - A) 1
 - B) 4
 - C) 2
 - D) 3
16. What is the primary function of a flip-flop in digital electronics?
 - A) To perform arithmetic operations
 - B) To generate clock signals
 - C) To store binary information and provide sequential logic
 - D) To perform logical AND operations
17. In a class AB push-pull amplifier, when does one transistor conduct while the other is off?
 - A) Both transistors are always conducting
 - B) One conducts during the positive half of the signal, and the other conducts during the negative half.
 - C) They conduct simultaneously.
 - D) They never conduct.
18. What is the main advantage of using an H-bridge circuit for motor control?
 - A) Low power consumption
 - B) Reduced complexity
 - C) Limited control options
 - D) High efficiency
19. What is the primary function of a Boost regulator in power electronics?
 - A) To increase the output voltage
 - B) To perform logical operations
 - C) To store data
 - D) To regulate the output voltage lower than the input voltage
20. What is the primary function of a Buck-Boost regulator in power electronics?
 - A) To increase the output voltage
 - B) To regulate the output voltage lower than the input voltage
 - C) To store data
 - D) To perform both step-up and step-down voltage conversion

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the application and working principle of SAR ADC in detail.	8	L*2	1	1
	b) Elaborate on the need for Sample and Hold Circuits. Also Briefly mention the working of it.	8	L2	1	1
2.	a) How is 555 timer used for different applications? Explain any 2 applications.	8	L2	1	1
	b) Explain the operation of an inverting Schmitt Trigger.	8	L2	1	1
3.	a) What are the different types of DACs. Explain any one in detail.	8	L2	2	1
	b) State the ideal characteristics of op-Amps.	8	L2	2	1

Unit – II

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|-------|--|---|----|---|---|
| 4. a) | Reduce the following Boolean expression using K-map:
$F(A, B, C, D) = \sum m(4, 5, 7, 8, 10, 11, 13, 14) + \sum d(0, 1, 2)$. | 8 | L2 | 3 | 1 |
| b) | Obtain the canonical product of the sum form of the following function:
$F(A, B, C) = (A + B')(B + C)(A + C')$. | 8 | L2 | 3 | 1 |
| 5. a) | Implement 1:16 Demultiplexers using ONLY 1:4 Multiplexers. | 8 | L2 | 4 | 1 |
| b) | Differentiate between a Latch and a FlipFlop. | 8 | L2 | 4 | 1 |
| 6. a) | Construct 4 to 16 Decoder using 3:8 Decoder. | 8 | L2 | 4 | 1 |
| b) | Derive the excitation table for SR Flipflop? How is SR converted to D Flipflop? | 8 | L2 | 4 | 1 |

Unit – III

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|-------|---|---|----|---|---|
| 7. a) | What is an H-Bridge? How this can be used for power amplification? | 8 | L2 | 5 | 1 |
| b) | What is a Power Amplifier? What are the types of power amplifier? | 8 | L2 | 5 | 1 |
| 8. a) | Elaborate on the basic role and working characteristics of a Switching Regulator. | 8 | L2 | 5 | 1 |
| b) | With its application, explain the working of Buck Boost Regulators. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
